F79SP Session 21: Poisson processes: toward applications

We discuss some extensions/modifications from homogeneous Poisson processes for more realistic applications.

Non-homogeneous Poisson processes

Here, we consider Poisson processes in which events or points occur at a *time-dependent* rate. The rate of the underlying Poisson process to be a function of t, $\lambda(t)$ instead of a constant λ.

Definition. A *(heterogeneous or non-homogeneous) Poisson process* with *rate function* $\lambda(t)$ is again a point process $N(\cdot)$ such that:

(PP3(i)) For any t and $h > 0$ (h small),

$$\begin{aligned} \mathbf{P}(N[t, t+h] = 0) &= && \text{as } h \to 0, \\ \mathbf{P}(N[t, t+h] = 1) &= && \text{as } h \to 0, \\ \mathbf{P}(N[t, t+h] \geq 2) &= && \text{as } h \to 0. \end{aligned}$$

(PP3(ii)) The numbers of points $N(I_1), \ldots, N(I_k)$ occurring in *disjoint* finite intervals $I_1, \ldots, I_k$ are *independent* random variables.

With this, properties of non-homogeneous Poisson process can be modified from homogeneous PP accordingly. For instance:

- For any finite interval $I = [t_1, t_2]$, the random variable $N[t_1, t_2]$ has a Poisson distribution with mean of:
- We can generalise Superposition and Splitting Theorem accordingly.
- However, that there is no straightforward alternative characterisation of a non-homogeneous Poisson process via exponential inter-arrival times.

Poisson processes in higher dimensions

Poisson processes can be defined in higher dimensions, for example two-dimensional spaces.

Compound Poisson process

Poisson processes are commonly viewed as processes of arrivals. In some applications, it is useful for us to further consider the weight/size of each arrival. We consider the following set up.

- Let $N(t), t \geq 0$ be a time-homogeneous Poisson process with rate λ.
- Let $\{X_n\}$ be an i.i.d. sequence of random variables which does not depend on the Poisson process.
- Assume that $a = \mathbf{E}X_1$ and $b = \mathbf{E}X_1^2$ are finite. Let also $\sigma^2 = \mathbf{Var}X_1$. Then $b = \sigma^2 + a^2$.
- Let $W(t) = \sum_{i=1}^{N(t)} X_i$.

Remark: For instance, in an insurance company setting:

- $N(t)$ represents the process of claims arrivals
- X_n is the size of the nth claim.
- $W(t)$ is the total of all claims within a time interval of length t.

Lemma 0.1. *The mean and the variance of $W(t)$ are given by*

$$\mathbf{E}W(t) = a\lambda t \quad \textit{and} \quad \mathbf{Var}W(t) = b\lambda t.$$

Proof. Without loss of generality, we may consider time interval $(0, t)$. We have $W(t) = \sum_{i=1}^{N(t)} X_i$. Then

$$\mathbf{E}W(t) \quad = \quad \sum_{n\geq 1} \mathbf{P}(N(t) = n)\mathbf{E}\left(\sum_{i=1}^{N(t)} X_i \mid N(t) = n\right)$$

and

$$\begin{aligned}\mathbf{E}W^2(t) &= \mathbf{E}\left(\sum_{i=1}^{N(t)} X_i\right)^2 \\ &= \sum_{n\geq 1}\mathbf{P}(N(t)=n)\mathbf{E}\left(\sum_{i=1}^{n} X_i\right)^2\end{aligned}$$

Then

$$\mathbf{Var}W(t) = \mathbf{E}W^2(t) - (\mathbf{E}W(t))^2 = (a^2+\sigma^2)\lambda t = b\lambda t.$$

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